

Performance Metrics, Position Sizing & Cost Models

A Quantitative Framework for Strategy Evaluation and Portfolio
Construction

TRUE ALPHA ACADEMY



Agenda: A Structured Journey Through Quantitative Strategy Analysis

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- 3 **Position Sizing** — Defining the problem and key frameworks
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- 6 **Transaction Cost Models** — Defining and quantifying market friction
- 7 **Optimal Sizing Problem** — Bringing it all together in a unified framework

Performance Metrics: The Quantitative Lens for Strategy Evaluation

What they are

Performance metrics are standardized, quantitative measures used to evaluate the risk-adjusted return profile of a trading strategy or portfolio. They translate raw P&L data into actionable, comparable signals.

Why they matter

Without rigorous metrics, strategy comparison is subjective and unreliable. Metrics allow practitioners to distinguish genuine alpha from noise, assess consistency, and benchmark against alternatives.

Metric	What It Measures
Cumulative Return	Total wealth growth over the period
Average Return	Typical periodic return (mean)
Volatility	Dispersion of returns (risk proxy)
Sharpe Ratio	Return per unit of risk
Turnover	Portfolio churn and implied cost

Cumulative vs. Average Return: Two Lenses on the Same Strategy

Cumulative Return

Total compounded gain over the evaluation horizon.
Answers: "How much did \$1 grow to?"

$$R_{\text{cum}} = \prod (1 + r_t) - 1$$

Average Return

Mean return per period. Answers: "What is the typical period return?"

$$\bar{r} = (1 / T) \sum r_t$$

Why both matter

- A strategy with a **high average return** but **high volatility** can still show weak cumulative performance due to variance drag.
- Cumulative return captures path dependency and compounding that the average misses.
- Average return is useful for period-by-period comparison and regime analysis.
- Together, they show whether returns are consistent or driven by a few outliers. **Median return** can help as it is more robust to outliers.

Volatility: Quantifying the Price of Uncertainty

Definition

Volatility is the annualized standard deviation of periodic returns. It is the most fundamental measure of risk in quantitative finance.

$$\sigma = \sqrt{[(252 / T) \times \Sigma (r_t - \bar{r})^2]}$$

Why it matters

- It is the denominator of nearly every risk-adjusted performance ratio.
- High volatility implies a wider distribution of outcomes — strategies may look profitable on average but be practically unacceptable due to drawdown risk.
- Directly impacts position sizing: a more volatile asset requires smaller allocations to maintain a target risk level.
- Realized volatility feeds into cost models: high-vol environments typically imply higher market impact.

Key insight: Volatility is not inherently bad — it is the *uncompensated* volatility that destroys value.

Sharpe Ratio: The Universal Benchmark for Risk-Adjusted Performance

Definition

The Sharpe Ratio measures the excess return earned per unit of total risk (volatility), relative to a risk-free rate.

$$\text{Sharpe} = (\bar{r} - r_f) / \sigma$$

Why it matters

- It is the most widely used metric for comparing strategies across asset classes and time horizons.
- A Sharpe > 1.0 is generally considered acceptable; > 2.0 is strong in a live trading context.
- It penalizes strategies that achieve high returns through excessive risk-taking.
- The Sharpe Ratio is the foundation for more advanced ratios (Sortino, Calmar, Information Ratio).

Important caveats

- Assumes normally distributed returns — can be misleading for strategies with fat tails or skewness.
- Sensitive to the choice of risk-free rate and evaluation period.
- A high in-sample Sharpe may not persist out-of-sample (overfitting risk).

The Sharpe Ratio's Blind Spot: Why It's Insufficient

The Problem

The standard Sharpe Ratio assumes returns are normally distributed and independent (i.i.d.). This can hide catastrophic tail risks.

- Can be "gamed" by strategies with negative skewness.
- Ignores the length of the track record.
- Fails to account for non-normality in returns.

Probabilistic Sharpe Ratio (PSR)

Accounts for skewness, kurtosis, and the sample size.

The Solution

PSR tells you the probability that the observed Sharpe is greater than a benchmark, adjusted for non-normality and track record length.

Turnover: The Hidden Cost Embedded in Every Strategy

Definition

Turnover measures the fraction of the portfolio that is rebalanced in a given period. It is typically expressed as an annualized percentage of AUM.

$$\text{Turnover} = \frac{1}{T} \sum_{t=1}^T \sum_i |w_{i,t} - w_{i,t-1}|$$

Practical Rule of Thumb

A 1% one-way transaction cost on a 200% turnover strategy reduces annual return by approximately 2% — enough to eliminate the alpha of many systematic strategies.

Why it matters

- Turnover is a direct proxy for transaction costs: higher turnover implies more frequent trading and greater friction.
- A strategy with a 200% annual turnover trades its entire book twice per year — costs compound rapidly.
- Turnover interacts with strategy capacity: high-turnover strategies are more sensitive to market impact at scale.
- It is a key input into cost-adjusted performance metrics and net Sharpe estimation.

ADVANCED METRICS

Beyond the Basics: Advanced Performance Metrics

Why go further: Core metrics provide a first-pass filter. Advanced metrics reveal the *quality* of returns — their consistency, tail behavior, and statistical robustness.

Cost Metrics

Turnover, Implementation Shortfall, Cost-Adjusted Sharpe

Trade Statistics

Hit Ratio, Win/Loss Ratio, Risk-Reward Ratio, Profit Factor

Advanced Ratios

Sortino Ratio, Calmar Ratio, Information Ratio, Omega Ratio

Statistical Tests

t-test on returns, Bootstrap Sharpe CI, White's Reality Check, Multiple Testing Correction

Key principle: *No single metric tells the full story. A robust strategy should score well across multiple dimensions simultaneously.*

Advanced Metrics in Detail: From Trade Statistics to Statistical Rigor

Trade Statistics

Hit Ratio

Fraction of trades with positive P&L — a measure of directional accuracy.

Win/Loss Ratio

Average winning trade divided by average losing trade — measures payoff asymmetry.

Risk-Reward Ratio

Expected gain per unit of expected loss — combines hit ratio and win/loss ratio.

Advanced Ratios

Sortino Ratio

Like Sharpe, but penalizes only *downside* volatility — more appropriate for asymmetric return distributions.

Calmar Ratio

Annualized return divided by maximum drawdown — captures tail risk and recovery.

Information Ratio

Excess return over a benchmark divided by tracking error — key for relative-return strategies.

Statistical Tests

Bootstrap Sharpe CI

Confidence interval for the Sharpe Ratio via resampling — guards against overfitting.

Henriksson-Merton Test

Computes the p-value of market timing for a specific signal/strategy.

Position Sizing: Translating Signals into Capital Allocation

The core problem

Given a set of return forecasts (alpha signals) and a risk model, how should capital be allocated across assets to maximize risk-adjusted returns?

$$\max_w f(w, \mu, \Sigma) \text{ subject to } g(w) \leq c$$

Where w is the vector of portfolio weights, μ is the expected return vector, Σ is the covariance matrix, and $g(w)$ represents constraints.

Why it is non-trivial

- Signal quality varies across assets and time.
- Assets are correlated — naive equal weighting ignores this.
- Risk is not static: volatility and correlations change.
- Transaction costs penalize frequent rebalancing — optimal sizing must account for friction.

The position sizing decision is where alpha is preserved or destroyed.

Diversification & Risk Models: From Markowitz to Modern Frameworks

Diversification Defined

Diversification allocates capital across assets so idiosyncratic risks cancel out, reducing portfolio volatility without proportionally reducing expected return.

$$\sigma_p^2 = W^T \Sigma W \Rightarrow \sigma_p < \sum_i W_i \sigma_i$$

(when correlations < 1)

Markowitz Mean-Variance (1952)

- Chooses portfolios on the efficient frontier for a given risk.
- Sensitive to estimation error in returns (μ) and covariance (Σ).
- Can produce concentrated, unstable portfolios in practice.

Modern Frameworks

Volatility Targeting

Scale positions to maintain target portfolio volatility.

Risk Parity

Equalize risk contribution

Kelly Criterion

Black-Litterman

Volatility Targeting: Maintaining a Constant Risk Budget

The Core Concept

Volatility targeting scales position sizes inversely to realized volatility. This ensures the portfolio maintains a target risk level, regardless of market regimes.

- Prevents "risk explosion" during high-volatility market crashes.
- Keeps the strategy "at work" during low-volatility periods.
- Foundational for systematic macro and trend-following.

$$w_i = (\sigma^* / \sigma_i) \cdot (1 / N)$$

Where σ^* is the target volatility and σ_i is the realized volatility of asset i .

Key Insight

By targeting risk rather than capital, you achieve a more stable return profile and improve long-term compounding.

Risk Parity: Equalizing Risk Contribution

The Core Concept

Allocate capital such that each asset or risk factor contributes equally to the total portfolio volatility. This balances the risk, not just the capital.

- Prevents concentration in high-volatility assets like stocks.
- Increases exposure to low-volatility assets like bonds.
- Produces a more resilient portfolio across different economic environments.

$$w_i \sigma_i = w_j \sigma_j$$

Where w_i is the weight of asset i and σ_i is its risk contribution.

Key Insight

True diversification is about balancing risk, not just dollars. In an equal-weighted stock/bond portfolio, stocks often contribute >90% of the risk.

Kelly Criterion: Maximizing Long-Run Wealth

The Core Concept

The Kelly Criterion determines the optimal size of a series of bets to maximize the long-term growth of capital. It provides the upper bound for rational position sizing.

- Maximizes the long-run geometric growth rate.
- Accounts for the probability of a win and the payoff odds.
- Foundational for gambling and modern quantitative finance.

$$w^* = (\mu - r_f) / \sigma^2$$

Where μ is the expected return, r_f is the risk-free rate, and σ^2 is the variance of returns.

Practical Application

"Full Kelly" is often too aggressive and leads to extreme drawdowns. Professionals typically use "Fractional Kelly" (e.g., 0.5x) to account for estimation error.

Black-Litterman: Combining Market Equilibrium with Investor Views

The Core Concept

A sophisticated asset allocation model that starts with the "market-neutral" portfolio (CAPM) and tilts it based on specific investor views.

- Uses a Bayesian framework to combine prior market info with subjective forecasts.
- Solves the "error maximization" problem of standard Mean-Variance Optimization.
- Produces more stable and intuitive portfolio weights.

$$P\mu = Q + \varepsilon$$

Where P represents the views, μ the returns, Q the view values, and ε the uncertainty.

Key Insight

Black-Litterman allows quants to blend systematic signals with market consensus in a mathematically rigorous way, leading to more robust portfolios.

Entropy Pooling: Beyond the Normal Distribution

The Core Concept

A flexible framework for portfolio construction that allows for "views" on any distribution, not just the mean and variance. It moves beyond parametric assumptions.

- Uses the Principle of Maximum Entropy to pool information from various sources.
- Handles non-normal returns, fat tails, and complex dependencies.
- Excellent for stress testing and non-Gaussian multi-asset allocation.

$$\max H(p) = - \sum p_i \log(p_i)$$

Where $H(p)$ is the entropy of the probability distribution p .

Key Insight

Entropy Pooling is the state-of-the-art for multi-asset allocation in non-Gaussian worlds, providing a robust way to handle real-world market complexity.

Transaction Costs & Cost Models: Friction Is Alpha's Greatest Enemy

Types of transaction costs

Cost Type	Description
Commission	Fixed or percentage fee paid to broker per trade
Bid-Ask Spread	Half-spread paid on each side of a round-trip trade
Market Impact	Price moves adversely as order is executed (temporary + permanent)
Slippage	Difference between expected and actual execution price
Opportunity Cost	Cost of not trading (missed alpha due to delay or partial fill)

Why cost modeling matters

- A strategy with a gross Sharpe of 1.5 and 300% annual turnover may have a *net* Sharpe below 0.5 after realistic costs.
- Cost models allow pre-trade estimation of net performance, enabling better strategy selection and sizing decisions.
- They inform the optimal rebalancing frequency – the trade-off between signal freshness and friction.

Models covered next:

Linear cost model, Quadratic (market impact) model, Implementation Shortfall model

Transaction Cost Models: The Linear Model

The Core Concept

Costs that scale linearly with the size of the trade. This is the simplest and most common way to model transaction costs for liquid assets.

- Commissions (fixed \$ per share or %).
- Bid-Ask Spread (half-spread paid on every trade).
- Best for small trades where your order doesn't move the price.

$$\text{Cost} = \sum |w_i - w_{i,\text{prev}}| \cdot (C + S/2)$$

Where C is the commission and S is the bid-ask spread.

Key Insight

The linear model is the "floor" of transaction costs; it represents the minimum friction you must overcome to be profitable.

Transaction Cost Models: The Quadratic (Market Impact) Model

The Core Concept

Costs that increase more than linearly as trade size grows relative to market liquidity. This models the "temporary impact" of eating through the limit order book.

- Large orders move the price adversely during execution.
- Impact is typically proportional to asset volatility.
- Primary constraint on the capacity of a trading strategy.

$$\text{Impact} \propto \sigma \cdot (\text{Trade Size} / \text{ADV})^\alpha$$

Where σ is volatility, ADV is average daily volume, and $\alpha \approx 0.5$ or 1.0.

Key Insight

The quadratic term penalizes large, fast rebalancing. It forces the optimizer to trade more slowly and limits the effective size of the strategy.

Transaction Cost Models: Implementation Shortfall (IS)

The Core Concept

Implementation Shortfall is the total cost of executing a trade, measured as the difference between the decision price and the final execution price.

- Explicit Costs (Commissions, taxes).
- Delay Costs (Price move between decision and order entry).
- Market Impact (Price move during execution).
- Opportunity Costs (Alpha lost on unfilled shares).

$$IS = P_{\text{Execution}} - P_{\text{Decision}}$$

Captures the total "leakage" of alpha during the entire execution lifecycle.

Key Insight

Implementation Shortfall is the gold standard for measuring execution quality. It ensures quants account for all frictions, not just the visible ones.

Optimal Sizing Under Costs: Bringing the Framework Together

Problem statement

Find portfolio weights w that maximize risk-adjusted net returns, accounting for signal quality, portfolio risk, and transaction costs:

$$w^* = \operatorname{argmax}_w \left[w^T \mu - (\lambda / 2) w^T \Sigma w - \kappa \cdot \text{TC}(w, w_{\text{prev}}) \right]$$

Where λ controls risk aversion, κ scales cost sensitivity, and $\text{TC}(\cdot)$ is the transaction cost function.

Key trade-offs encoded

- **Alpha vs. Risk:** Higher expected return must justify higher portfolio variance.
- **Alpha vs. Cost:** Rebalancing toward the optimal is only worthwhile if the gain exceeds the friction.
- **Turnover Penalty:** The cost term naturally limits excessive rebalancing — the model self-regulates.

Implementation

A companion Jupyter Notebook provides a compact Python implementation, including:

- Covariance estimation
- Cost model calibration
- Efficient frontier visualization

This is where theory meets practice.